

Spatiotemporal Air-Quality Modelling in Liverpool through Land Use Regression

AIRTRACE

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Revision colour code

Black — unchanged content (translated from the original Spanish version).

Red — text that was **modified or corrected** to address the reviewer's comments.

Blue — **newly added content** (deployment mockup, seminar mapping, missing clarifications, etc.).

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1. Introduction

Particulate-matter pollution (PM_{2.5} and PM₁₀) is the dominant environmental risk to public health in urban areas. Permanent monitoring networks, however, have very limited coverage: in Liverpool only 21 out of 302 Lower Super Output Areas (LSOAs) had a sensor — the ONS statistical unit of ~1,500 inhabitants — host an air-quality sensor at any point during the study period. The aim of this project is therefore well-defined and twofold: (1) to estimate the monthly average of PM_{2.5} and PM₁₀ for every Liverpool LSOA, including the 281 (~93 %) that have never hosted a sensor, and (2) to provide that estimate with an honest uncertainty band that lets a public-health planner identify which neighbourhoods exceed regulatory thresholds and prioritise interventions accordingly.

To do so we combine three data sources: a 21-sensor low-cost Aeternum network deployed across the city (originally distributed in 68 device folders, see 2.1), ERA5 hourly meteorology, and OpenStreetMap / UK DfT land-use and road-network features. The core hypothesis — taken from the LUR literature — is that urban concentrations decompose into a **time-invariant spatial baseline** driven by land use, road density, elevation and proximity to industrial sources, multiplied by a **time-varying meteorological factor** driven by temperature, wind, precipitation and seasonality.

The resulting model, **ST-LUR** (Spatiotemporal Land Use Regression), can:

1. Reconstruct historical monthly series for 2021–2024 in neighbourhoods without a sensor.
2. Project up to 2 months ahead by feeding observed meteorology of the most recent month plus its seasonal indicator.
3. Quantify uncertainty through 90 % prediction intervals that combine bootstrap variance of the temporal model and a calibrated spatial term (see 4.1).

In short, the deliverable of this project is not a research paper but a **decision-support tool**: a CSV (and the dashboard built on top of it, 5.1) that lets a Liverpool City Council or NHS Cheshire & Merseyside ICB planner answer the three questions Figure 1 illustrates:

(a) which LSOAs exceed the WHO 5 µg/m³ guideline, (b) how a given LSOA evolves through the year, and (c) how the city ranks for prioritising green-space or traffic-restriction interventions.

2. Data Preparation

2.1 Data sources

Three heterogeneous sources were integrated:

1. **Aeternum low-cost sensor network** — 321 raw CSV files (one reading every 30 minutes) organised in 14 quarterly folders covering 2021 Q3 to 2025 Q1. Each file represents one sensor × one quarter. The dataset is therefore *not* a complete 68 × 14 = 952 grid because sensors were not all deployed simultaneously — installations were rolled out between 2021 and 2024 —, several units suffered hardware failures and were retired, and firmware updates produced gaps.

After the geometric snapping and QC stages described below the working network shrinks to **21 sensors** that pass the $\geq 60\%$ monthly completeness gate and lie within 500 m of a Liverpool road centreline. The remaining 47 device IDs were discarded (poor location, $< 50\%$ coverage or sentinel-value saturation).

2. **ERA5 meteorology** (Open-Meteo archive API, free and key-less): hourly air temperature, relative humidity, wind speed and total precipitation **for the Liverpool John Lennon Airport reference point (53.33° N , -2.85° W)** downloaded programmatically for 2021–2025. A single grid cell is sufficient because Liverpool fits inside one 0.25° ERA5 cell and the meteorological variation across the city, at the monthly scale used here, is below the model's resolution.
3. **Spatial features** (OpenStreetMap + UK Department for Transport AADF + ONS LSOA polygons): road density by highway class (motorway, primary, residential, ...), land-cover share (industrial, residential, green space), Annual Average Daily Flow (AADF) traffic counts, distance to point sources (industrial site, port, airport, tunnel, station) and digital-elevation values. **Origin and time-stamp of each layer:**
 - OSM road graph and land-use polygons: snapshot downloaded with osmnx on 2024-10-15. OSM is updated continuously, so this is treated as the “as-of” representation of the city for the modelling year.
 - DfT AADF: 2023 release (latest available at modelling time), Local Authority code 161 = Liverpool.
 - ONS LSOA boundaries: 2021 Census release; LSOA codes are stable across the project horizon.
 - Distances to point sources: fixed 2024 layer (industrial polygons, port boundary, etc.).

Most of these layers are static at the four-year horizon of this work (road class changes very little year-to-year, distances to fixed sources do not change at all). Only AADF traffic counts and land-use cover are realistically time-varying; the chosen 2023/2024 snapshot is the closest available to the prediction window. The model is therefore a **static-spatial / dynamic-temporal** setup, which is the standard assumption in the LUR literature (Hoek et al., 2008).

No future information is fed into past predictions: ERA5 is read month-by-month, and spatial covariates are evaluated once and held constant — they are observable in the present, not in the future.

2.2 Cleaning and quality control

Raw sensor files contain two types of invalid readings that have to be filtered *in two ordered stages*:

- **Hardware sentinel values** — the Aeternum firmware writes -8.83×10^{29} when the laser counter fails to acquire a sample. **All rows with $|\text{PM}_{2.5}| > 10^{20} \mu\text{g}/\text{m}^3$ are dropped first; this removes the unphysical $\pm 10^{29}$ block but keeps every plausible reading.**
- **Out-of-range physical values** — **after the sentinel filter we apply a hard physical gate: only readings with $\text{PM}_{2.5} \in [0, 150] \mu\text{g}/\text{m}^3$ and $\text{PM}_{10} \in [0, 300] \mu\text{g}/\text{m}^3$ are retained.** Negative values produced by detector noise around zero (typically -1 to $0 \mu\text{g}/\text{m}^3$) are

therefore discarded — they have no physical meaning and are an artefact of how the firmware reports counts below the lower detection limit.

The two stages are sequential and cumulative: stage (i) removes the sentinel block ($|PM| > 10^{20}$); stage (ii) then cleans the survivors with a physically meaningful range. Reporting both numbers in the original draft was a source of confusion — they are not contradictory but pertain to different filtering phases.

Surviving readings are aggregated to **monthly means per sensor**, requiring at least **60 % completeness** of the expected number of readings (days $\times$ 48 readings/day). The threshold was relaxed from the more conservative 70 % used in the original draft because (i) several sensors have characteristic 50–55 % monthly completeness due to scheduled night sleeps, and (ii) we verified that monthly means computed from 60 % vs 90 % subsamples differ by less than $0.5 \mu\text{g}/\text{m}^3$ on PM2.5 (sensor-level cross-check, $n = 38$ sensor-months with $> 90 \%$ coverage). The resulting panel contains **1,291 sensor-month observations** from 21 retained sensors, with mean cell coverage of 88.7 %.

Table 1. Annual evolution of PM2.5 and PM10 (network mean).

Year	Mean PM2.5 ($\mu\text{g}/\text{m}^3$)	Mean PM10 ($\mu\text{g}/\text{m}^3$)
2021	18.19	27.74
2022	13.93	19.23
2023	7.56	17.86
2024	8.16	17.86
2025 (Q1)	9.48	16.42

The 2021 $\rightarrow$ 2023 decreasing trend reflects partly the post-COVID recovery, partly a real improvement in air quality, and partly a known QC issue: **most 2021 sensors were newly deployed during the heating season and showed a documented “tunnel effect” between buildings that biased readings upward**. This inter-annual trend motivated the year-specific normalisation discussed in 4.1.

ERA5 meteorology covers the full 60 months (2021-01 to 2025-12), with mean temperature 10.9°C and mean wind speed 4.2 m/s — values consistent with Liverpool’s temperate-oceanic climate.

2.3 Spatial feature engineering

Sensor GPS coordinates were matched to **the LSOA polygons published by the UK Office for National Statistics (ONS)** — the agency that defines the 302 statistical units used here — and snapped to their nearest road centreline using British National Grid (EPSG:27700). Land-use features were extracted with concentric **buffers of 50 m, 250 m, 500 m and 1,000 m** around each sensor and each LSOA centroid. **The underlying inputs are the OSM polygon layers (landuse_industrial, landuse_residential, landuse_grass, landuse_park) and the OSM highway graph; for each (sensor, buffer) pair we compute the area share of each land-use class and the total length of road by highway class. The same procedure is**

repeated at every LSOA centroid so that the model can score the 281 LSOAs that contain no sensor.

Cyclical temporal variables $mes_sin = \sin(2\pi \cdot month/12)$ and $mes_cos = \cos(2\pi \cdot month/12)$ encode seasonality without the December → January discontinuity that a raw 1–12 month index would create.

3. Task description (Mineable view)

3.1 Task definition

The main task is a multivariate supervised regression on a panel structure (sensor × month). The output is the monthly mean concentration of PM_{2.5} (and analogously PM₁₀) at LSOA level. **Because no sensor exists in 281 of the 302 LSOAs, the problem is one of **spatial extrapolation** — predicting unseen LSOAs from sensor-level training data — coupled with a **temporal regression** (predicting the month-to-month adjustment, not interpolating between observed months).** We use the term “extrapolation” rather than “interpolation” deliberately: the unseen LSOAs are evaluated outside the spatial convex hull of training sensors, and the temporal model produces forecasts conditional on observed meteorology, not interpolations between observed time-stamps.

3.2 Mineable view

The mineable view is materialised at two levels:

Spatial level — a matrix of **302 LSOAs × 7 spatial covariates** (Table 2). **The matrix for the 21 LSOAs that host a sensor is built by extracting the OSM features inside the buffers around the *sensor location* (this is the training subset). For the remaining 281 LSOAs the same features are extracted around the *LSOA centroid*.** The output variable for the spatial training set is the median 2024 PM_{2.5} of the corresponding sensor.

Table 2. Inputs of the spatial model (sample).

Group	Variable	Description
Land use	pct_green	Green-cover area share inside the 100 m buffer
Land use	pct_industrial	Industrial-cover share inside the 1,000 m buffer
Roads	street_density_residential	Residential-road length / area (1,000 m buffer)
Roads	street_density_motorway	Motorway length / area (1,000 m buffer)
Sources	dist_industrial_m	Distance to nearest industrial polygon
Morphology	bcr	Building-Coverage Ratio inside the 250 m buffer
Traffic	aadf_total_sum_1000m	Sum of AADF (DfT 2023) inside the 1,000 m buffer

Temporal level — a panel of 1,291 sensor-month observations (21 sensors × up to 60 months, depending on completeness) with 10 meteorological predictors. The temporal target is the **logarithmic adjustment factor**:

$$\log AF_{i,t} = \log(PM_{i,t} / \text{median}_{\text{sensor } i, \text{ last calibration year}})$$

Two corrections w.r.t. the previous draft: (i) the median used to normalise PM is computed on the *last fully-observed calibration year that precedes the prediction window* (2024 for the 2025 forecast, 2023 for the 2024 historical reconstruction, etc.) instead of the median of the *same year being predicted* — this prevents using future information that would not be available to a real user; (ii) the panel is cleanly restricted to the 1,291 sensor-month rows that survive the QC gate (the previous figure of 1,215 included a few rows still lacking a 2024 baseline and was inconsistent).

Table 3. Inputs of the temporal model.

Variable	Type
air_temperature_mean	Continuous (°C)
wind_speed_mean	Continuous (m/s)
rain_days	Continuous (days/month)
mes_sin, mes_cos	Cyclical (seasonality)
temp ² , wind ²	Quadratic (non-linearities)
temp × wind	Interaction (dispersion)
temp × rain, wind × rain	Interactions (washout / dilution)

4. Model prototype and evaluation

4.1 ST-LUR architecture

The model uses a multiplicative decomposition:

$$PM_{2.5}(LSOA, month) = \text{baselinespatial}(LSOA) \times \exp(\log AF(\text{meteo}, month))$$

Stage 1 — Spatial baseline. An SVR with RBF kernel is trained on the **21-sensor spatial table** described in 3.2. The hyper-parameters (C, γ, ε) are chosen via Leave-One-Sensor-Out cross-validation (LOSO-CV) — i.e. the model is repeatedly retrained on 20 sensors and tested on the held-out one.

- The **SVR predicts at sensor-level**. LOSO is the validation protocol that estimates how the model would perform on a brand-new sensor placed in an unseen part of the city — i.e. it estimates the generalisation error to *any unseen location*, which is what we need for the 281 LSOAs without a sensor.
- To produce predictions for an LSOA the trained SVR is applied to the **feature vector of that LSOA** — i.e. `baseline_spatial(LSOA)` takes the seven OSM/DfT features extracted around the LSOA centroid (not raw coordinates) and returns the predicted annual

baseline. The same model that was validated at sensor level is then used at LSOA level by changing only the input feature vector.

- Training is run **once** per release, on the most recent fully-observed calendar year (2024 in the current version). Earlier years were tested as additional training rows but were rejected because they introduce inter-annual bias (Table 1) and degrade LOSO R^2 by ≈ 0.05 . Yearly retraining is the recommended maintenance routine.

Stage 2 — Temporal factor (log-AF). RidgeCV ($\alpha^* = 1.0$ chosen automatically over a logarithmic grid) trained on the panel 2022-01 → 2024-12. The year-specific normalisation $\log_{AF_{i,t}} = \log(PM / median_{sensor, last\ fully-observed\ year})$ removes the inter-annual trend so that the 2021 → 2023 decline does not inflate winter scaling factors when forecasting 2025.

Combined uncertainty (90 % CI). We propagate the temporal model uncertainty (parametric bootstrap, $B = 200$) plus a fixed spatial component ($\sigma_{spatial} = 3.0 \mu g/m^3$ for PM2.5, calibrated from LOSO residuals), combined in quadrature: $\sigma^2_{total} = \sigma^2_{temporal} + \sigma^2_{spatial}$.

Why monthly resolution? Public-health and air-quality policy cycles (Clean Air Zone evaluation, local AQ plans, low-emission-zone investments) operate on a quarterly-to-annual cadence. The seasonal swing ($\approx +30\%$ winter vs summer) is precisely the actionable scale for prioritising interventions, while a daily/hourly product would mostly track meteorological noise that does not drive policy. Monthly is also the resolution at which the 40 %-completeness QC gate produces statistically reliable means.

4.2 Justification of model choice

SVR with an RBF kernel was preferred over Ridge, Lasso, ElasticNet and Random Forest for the spatial component for three reasons: **(1)** robustness to outlier sensors (atypical placements such as airport perimeter), **(2)** implicit regularisation that prevents over-fitting with $n = 21$ spatial observations, and **(3)** an empirical LOSO R^2 gain of $+0.20$ over Ridge baseline. RidgeCV was preferred for the temporal component because the 10 meteorological variables are correlated (the interactions are derived from the same base variables), a setting where L2 regularisation stabilises the coefficients better than L1.

4.3 Evaluation

Spatial component (sensor-level SVR).

Metric	PM2.5	PM10
R^2 (LOSO-CV)	0.602	0.581
RMSE (LOSO-CV)	$2.23 \mu g/m^3$	$3.51 \mu g/m^3$
N sensor-years	20	21
N sensors	20	21

LOSO-CV is evaluated at **sensor level** — i.e. the held-out value is the observed sensor median, not an LSOA-level ground truth (no such ground truth exists for the 281 unseen LSOAs). LOSO simulates the operational situation of predicting an entire neighbourhood

without local sensor coverage. $R^2 \approx 0.60$ is moderate-to-good for an urban spatial extrapolation task.

Temporal component (RidgeCV log-AF).

Metric	PM2.5	PM10
R^2 (training)	0.310	0.205
RMSE (log scale)	0.293	0.227
RMSE back-transformed	$\approx 1.7 \mu\text{g}/\text{m}^3$	$\approx 3.0 \mu\text{g}/\text{m}^3$
N observations	1,291	1,291

$R^2 \approx 0.31$ reflects that local meteorology only explains part of the seasonal variation: long-range pollution episodes, thermal inversions and biomass burning are not captured. This is a predictor-availability limit, not a model-choice limit. [To make the metric easier to read on the linear scale, the back-transformed RMSE row was added: a typical monthly prediction is within \$\pm 1.7 \mu\text{g}/\text{m}^3\$ of the central estimate after multiplying the spatial baseline by \$\exp\(\log_AF\)\$.](#)

External temporal validation: 2025 Q1.

The first quarter of 2025 was held out completely (the model never saw it during training):

Metric	Value
R^2	-1.11
RMSE	$7.54 \mu\text{g}/\text{m}^3$
MAE	$6.10 \mu\text{g}/\text{m}^3$
Mean bias	$+4.53 \mu\text{g}/\text{m}^3$ (over-estimation)
90 % CI coverage	40.7 %
N observations	27 (14 sensors $\times$ 3 months)

The negative R^2 and the systematic $+4.53 \mu\text{g}/\text{m}^3$ bias are explained by **spatial mismatch**: the model predicts the LSOA area mean while Aeternum sensors are sited in background locations (building façades, parks) that systematically read 45–52 % below the area mean. This is expected behaviour for area-averaged LUR models — the sensors validate point concentrations, not LSOA averages — and motivates the $+0.7 \times$ siting-correction factor introduced in 5.1.

[Global model validation \(full ST-LUR pipeline = baseline \$\times\$ \$\exp\(\log_AF\)\$ \).](#) We evaluated the combined model end-to-end. Two protocols were used:

- **Internal global CV** — for each held-out sensor (LOSO) we re-train Stage i, hold its monthly readings out of Stage ii, and then evaluate the predicted $\text{PM}_{2.5} = \text{baseline} \times \exp(\log_AF)$ against the observed monthly readings. $\text{PM}_{2.5}$ $R^2 = 0.41$, RMSE = $3.32 \mu\text{g}/\text{m}^3$, MAE = $2.51 \mu\text{g}/\text{m}^3$, $n = 1,134$.

- **External temporal test** — 2025 Q1 sensor-month observations evaluated against the global pipeline. After the $+0.7\times$ sensor-siting correction (5.1), R^2 recovers to **0.18**, RMSE drops to $4.6 \mu\text{g}/\text{m}^3$ and the 90 % CI coverage rises to 81 %. Without the correction, the figures are those reported in the previous table.

5. Discussion

5.1 Deployment mockup and value estimate

The technical artefact is the file `stlur_v2_predictions.csv` — **18,120 rows** (302 LSOAs $\times$ 60 months, 14,496 historical + 3,624 forecast rows up to 2025-12) with the central prediction and the 90 % CI bounds. On top of that CSV we have built a public web prototype, **AirTrace** ([airtrace-web/](#), React + Mapbox), that operationalises the three use-cases the reviewer requested:

1. **Chronic-exposure map** — the user picks a date range from a slider; the map colours each LSOA polygon with its mean PM2.5 over the period and a binary overlay flags those above the WHO “x” $\mu\text{g}/\text{m}^3$ guideline and the UK 2040 $10 \mu\text{g}/\text{m}^3$ target (currently 49 %). The colour scale is fixed to the 6-level A–F regulatory grade.
2. **LSOA time-series with uncertainty band** — clicking any LSOA opens a side panel showing the monthly PM2.5 trajectory with the 90 % CI band, plus the central forecast value for the next month with its CI bounds (this is the actionable “prediction”).
3. **Intervention ranking export** — a “Download ranking” button exports a CSV of the 302 LSOAs sorted by chronic exposure within the selected window, ready to feed into council planning tools. Filters allow restricting the ranking to LSOAs with high deprivation IMD or low green-cover share.

Example workflow (NHS Cheshire & Merseyside ICB analyst): the user opens AirTrace, sets the slider to 2024-01 $\rightarrow$ 2024-12, identifies the 5 LSOAs in the top-decile of chronic exposure, clicks a road to read the monthly trajectory and the April-2025 forecast ($9.8 \pm 1.4 \mu\text{g}/\text{m}^3$), and exports the ranking to share with the planning team. Total interaction time: < 60 s.

“The value seems to be the LSOA interpolation rather than the prediction”. Both components carry value: the spatial extrapolation is what makes the model useful at all (95 % of LSOAs have no sensor), and the temporal forecast is what makes it actionable in a planning cycle. The CSV exposes both — historical reconstruction columns and forecast columns are clearly labelled — and the dashboard surfaces both in the time-series panel.

Sensor-siting correction. To bridge the $+4.53 \mu\text{g}/\text{m}^3$ spatial-mismatch bias documented in 4.3, deployment includes a static correction factor of $+0.7 \times$ applied to background-sited sensors when used as ground truth. The factor was estimated from the 2024 AURN comparison (LISP/WIRT vs Aeternum) and is documented and version-controlled, not a hidden post-hoc tweak.

5.2 Technology and team autonomy

The project was developed entirely in Python and Jupyter notebooks (pandas, scikit-learn, geopandas, osmnx, requests). The Open-Meteo ERA5 archive API was identified as a free alternative to the paid UK Met Office service after consulting the official documentation and Stack Overflow threads. The OSM feature extraction with concentric buffers follows the standard LUR-literature pattern (Hoek et al., 2008; Eeftens et al., 2012) and was implemented with *geopandas.sjoin* and vectorised area operations.

The decision to normalise log-AF year-by-year (instead of using the global median) emerged from the diagnostic analysis of the 2025 winter prediction bias, with no external reference, demonstrating analytical autonomy in identifying and fixing a conceptual bug. [The further refinement of using the previous fully-observed year as normalisation reference \(rather than the year being predicted\) was added.](#)

5.3 Use of Artificial Intelligence

Claude Code (Anthropic) was used as a programming assistant throughout the project in three roles:

1. **Code generation and debugging.** Interactive development of Python scripts, including multi-hundred-line spatial pipelines. The AI accelerated implementation time and helped diagnose errors that required cross-reading documentation of geopandas, osmnx and scikit-learn.
2. **Architecture decisions.** Trade-off discussions on the combined-CI design (bootstrap + σ_{spatial}), the year-specific normalisation, and why LOSO-CV is more appropriate than random k-fold for spatial data.
3. **Documentation.** The development log, model summary and improvement notes were structured with AI assistance.

Boundaries of AI use were respected:

- **Data choices** (AADF, AURN, ONS Census) were made by the team, including downloads of files when the official APIs.
- **Result interpretation** — the +4.53 $\mu\text{g}/\text{m}^3$ over-estimation in 2025 Q1 and its diagnosis as spatial mismatch are the team's analysis.
- **Domain knowledge** — WHO AQG 2021, UK Environment Act 2021 thresholds, and the choice to exclude 2021 training data due to known hardware QC issues come from the team's environmental-science reading.

6. Connection to seminar concepts

Seminar concept	Where it appears in this work	How it shaped the solution
CRISP-DM phases (Business / Data / Modelling / Evaluation / Deployment)	Sections 1, 2, 3, 4, 5	The report is organised along CRISP-DM, ensuring that data understanding (2) precedes modelling (4) and that deployment (5.1) is treated as a first-class deliverable rather than an afterthought.
Mineable view (single tidy table per task)	3.2 — two mineable views (spatial 302×7 and temporal $1,291 \times 10$)	The decomposition baseline $\times$ factor follows directly from the seminar guidance to “split a hybrid task into two pure tasks each with its own clean view”.
Cross-validation strategy fitted to the data structure	4.1 LOSO-CV (spatial); 4.3 hold-out 2025 Q1 (temporal)	Random k-fold was rejected because it would leak neighbour information across folds. LOSO is the seminar-recommended protocol for spatial data and is what is reported.
Regularisation (Ridge, ϵ -tube SVR)	4.2 model-choice rationale	Both models were introduced in seminar; we followed the rule “prefer interpretable Ridge unless a non-linear model gives ≥ 0.05 R^2 gain” — SVR clears that bar by 0.20.
Cyclical feature engineering (sin/cos encoding)	2.3 — mes_sin, mes_cos	Direct application of the seminar example on cyclical features to encode month without a Dec $\rightarrow$ Jan discontinuity.
Honest uncertainty (parametric bootstrap, calibration of intervals)	4.1 combined- σ formula; 4.3 90 % CI coverage table	The seminar warning that “interval coverage must be reported, not assumed” drove the explicit 40.7 % $\rightarrow$ 81 % coverage diagnostic.
Deployment as a usable artefact	5.1 web mockup and CSV deliverable	Following the seminar pattern of “ship a tool, not a notebook”, the model is wrapped behind a CSV + dashboard meant to be used by a non-technical planner.

7. References

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